SOLARIS

Charity-Driven Off-Grid Energy Needs Forecaster for Unelectrified Communities in Developing Countries

Meet The Team…

Amara B

Project/Product Background
Ex Amazon, Ex Meta

Gary G

Machine Learning @ UCL

Rishi D

Computer Systems Engineering @ Brunel University

Viral C

Computer Science @ University of Manchester

Zohaib E

Computer Systems Engineering @ Brunel University

The Issue

Energy poverty + weak planning = stranded capital and slower impact

No Infrastructure

666M people lacked electricity access in 2023

Of these, around 565M are in sub-Saharan Africa (≈85% of the global deficit); Central and Southern Asia's gap has fallen to only about 27M.

Lack of Demand Planning

Manual planning and weather volatility lead to poor demand forecasts.

No use of currently AI to help assist with the demand planning

Global Missions

UN SDG - affordable, reliable, sustainable, modern energy for all by 2030.

World Bank Energy Access Compact - accelerating to 1.5–2% annual growth in low-income nations

Our Solution

Multi-agent AI on OpenClaw: Autonomous demand forecasting for unelectrified communitie

Satellite Imagery

Remote sensing - Night lights + land use (Sentinel/Planet data)

Weather + Demographics

Context signal - Gridded population

AI Agents

Forecasting + planning - Agentic LLM workflows

Actionable Plans

NGOs • UN • World Bank

Autonomous, climate-aware predictions to boost aid efficiency in unelectrified communities.

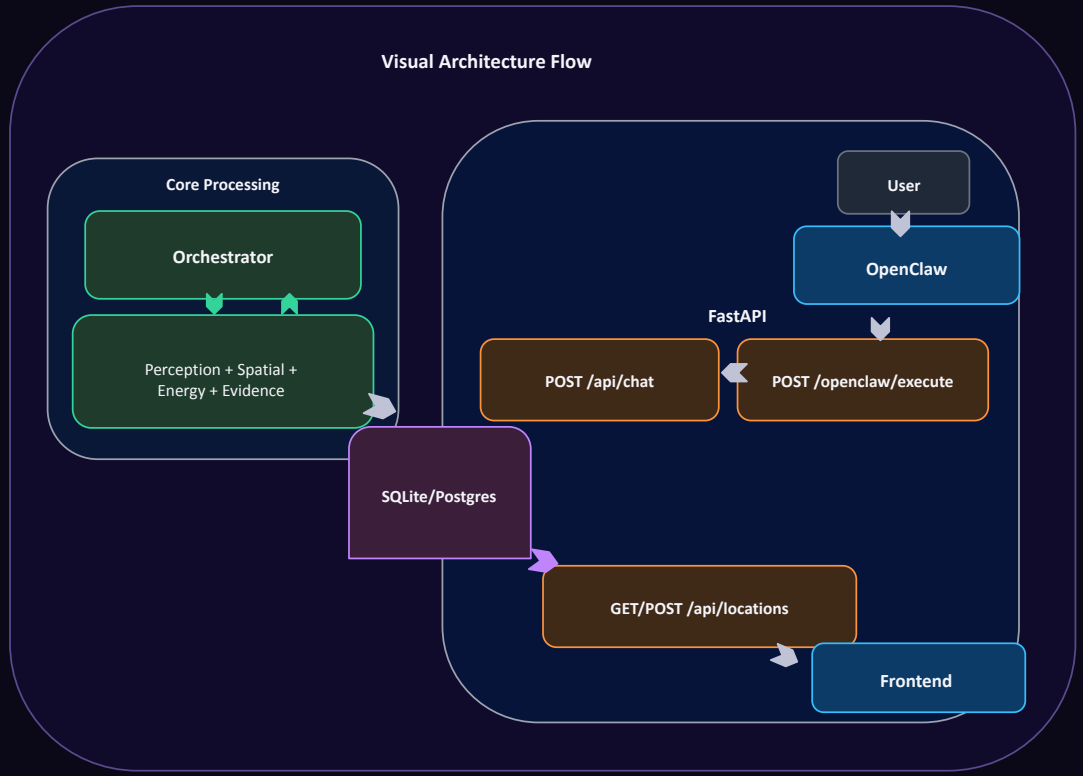
Data-driven Energy Allocation

More connections. Less waste. Higher impact

- NGOs planning village-scale energy projects
- UN agencies coordinating SDG programs
- World Bank / WEF funding off-grid interventions

- Better demand forecasts
- Faster, lower-waste deployment decisions
- Higher energy access impact per dollar

Workflow



Frontend Focus

- UX Layer: dashboard/map + forms
- Operator actions: input village data, run forecast, inspect confidence
- API consumer of /run endpoints
- Demo advantage: decision support visible in real-time

Tech Stack

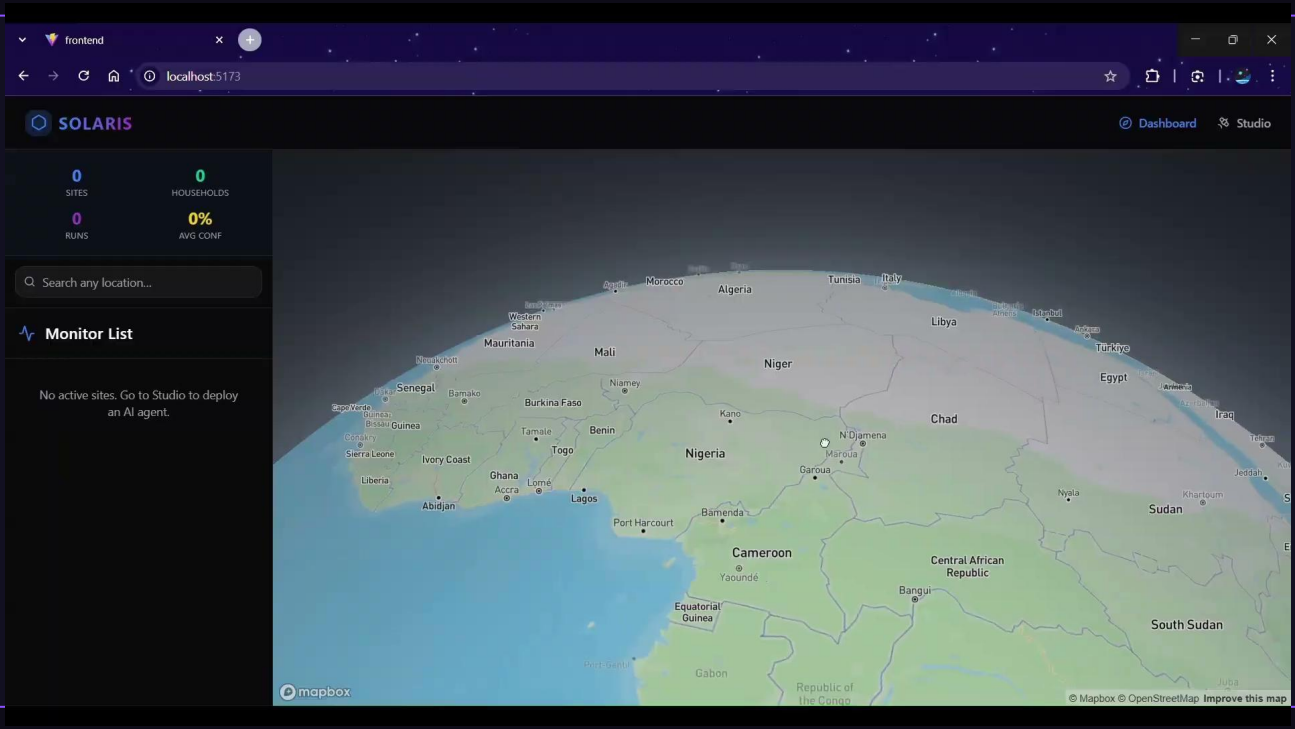
Backend: Python, FastAPI, Uvicorn, Pydantic
Data: PostgreSQL (psycopg), SQLite dev
AI Runtime: OpenClaw multi-agent pipeline
Quality: pytest, jsonschema, ruff
LLM Models: Qwen, GLM5

Reliability

VLM fallback + contract validation + optional API token auth

Demo

From raw signals to deployment intelligence



Evaluation & Future Work

Technical Evaluation
Validation + contracts + quality signals

System Reliability
Fallbacks + runtime observability

Beyond Energy
Reusable multi-agent planning engine

Technical Evaluation

- Validation coverage: unit + integration tests (pytest), API smoke tests, degraded-mode tests
- Contract checks: shared pipeline schema (pipeline.v1.json) + VLM contract validator
- End-to-end demo QA: run_demo_bundle.py + judge_run.py
- Quality telemetry exposed by API: confidence, fallback flags, runtime errors via /run/{id}/quality

Future Work + Platform Reuse

- Short-term: stronger benchmark metrics (forecast error by region/season), richer frontend explainability
- Scale path: replace deterministic blocks with learned models as data maturity increases
- Not limited to energy: same architecture can support healthcare resource forecasting, water access planning, disaster-response logistics, and agriculture input planning by swapping domain agents/objectives while keeping orchestration + API

Bounties We Are Submitting To

1) An AI Agent for satellite imagery analytic

Key Requirements

- Build an AI agent that detects real-world events
- Retrieve relevant satellite data (e.g., Sentinel-2)
- Analyze and deliver commercial insights

Why Solaris Fits

Solaris already combines geospatial context + analysis agents for actionable field recommendations.

Sponsor

The Compression Company

2) AI Agents for Good

Key Requirements

- OpenClaw-based autonomous agents
- SDG-aligned real-world impact
- FLock API + open-source models
- Multi-channel deployment

Why Solaris Fits

Solaris is, built on OpenClaw, and deployable via chat/API workflow.

Sponsor

Flock.Ai

3) Z.AI General bounty

Key Requirements

- Integrate Z.AI GLM models as core component
- 5–10 min pitch/demo video
- Public GitHub repo + clear README
- Working prototype (live demo preferred)

Why Solaris Fits

Solaris has a working multi-agent prototype and can add GLM-based components for this track.

Sponsor

Z.Ai

Thank You